

Training Program for Big Data & Distributed Systems

Bridge Role — Supporting Until Permanent Position

Structured, Intensive, Career-Level Training

This is a bridge role intended to support you until you secure a permanent position. The training path is structured and suggestive. Realistically, it takes 6 months to 2 years to gain full readiness for the role, depending on

Company QcFinance.In
Implementation for Joshi LLC NJ USA

Additional Learning Resources:

In addition to the Udemmy courses listed below, there are **YouTube playlists** based on the project that you must follow. These will be shared incrementally as you progress through each phase.

The courses must be taken in the order listed below, as each builds on the previous one. Training will focus heavily on **Python, distributed systems, testing, and big-data computation.**

Let me know once you start, and we will review progress weekly.

Your role might not be exactly what is in the course, but it will give you some idea of what skills are needed and how to approach them. As time goes by, the role might evolve and become more focused on Copilot and AI-based prompts and agents. However, the working style, methodology, and culture will remain the same.

Best regards,
Shivgan Joshi India MP Indore

Complete Course Links (Suggested Sequence)

- Course 1:** Note taking Course and Shadow Analyst for C Suite Analyst
<https://www.udemy.com/course/note-taking-course-and-shadow-analyst-for-c-suite-analyst/>
- Course 2:** Remote & Independent Working in Managerless Environments
<https://www.udemy.com/course/remote-independent-working-in-a-managerless-environment-101/>
- Course 3:** Connecting & Managing Ubuntu Bridge Computers for Remote Work
<https://www.udemy.com/course/connecting-managing-ubuntu-bridge-computers-for-remote-work/>
- Course 4:** Research & Writing Skills for Essays in Controlled Environment on Bridge Node
<https://www.udemy.com/course/research-writing-skills-for-essays-in-controlled-environment>
Below are Job specific, check with your project _____

- Course 5:** Big Data Infrastructure Administration — Ticket-Based Learning
<https://www.udemy.com/course/big-data-infra-admin-101-ticket-based-learning/>
- Course 6:** Python Full-Stack Computational Framework (Beginner)
<https://www.udemy.com/course/python-full-stack-computational-framework-for-beginners-101/>
- Course 7:** Full-Spectrum Comprehensive Testing & Error Reading
<https://www.udemy.com/course/full-spectrum-comprehensive-testing-error-reading-101/>
- Course 8:** Python Big Data Refresher — Intermediate (Quiz Based)
<https://www.udemy.com/course/refresher-python-big-data-intermediate-quiz-based-course-102/>
- Course 9:** Python Full-Stack Backend & ML Engines
<https://www.udemy.com/course/python-full-stack-and-backend-engines-for-mc-ml-engines-102/>
- Course 10:** Writing Tests for Simulation Engineering & Code Conversion
<https://www.udemy.com/course/writing-tests-for-simeng-python-code-conversion-concepts-101/>
- Course 11:** Full-Stack Computational Framework for Big Data Simulations
<https://www.udemy.com/course/full-stack-computational-framework-for-big-data-simulations/>
- Course 12:** Big Data Code Changes for Full-Stack Simulation Engine 2026
<https://www.udemy.com/course/big-data-code-changes-for-full-stack-simulation-engine-2026/>
- Course 13:** Converting Standalone Python Code to Distributed Analytics Code
<https://www.udemy.com/course/converting-standalone-code-to-distributed-code-for-analytics/>

Training Assumptions

- Each course requires approximately 2–4 days
- Average effort: 4–6 hours per day
- Training cadence: one course per week
- Initial readiness: approximately 13 weeks (all 13 courses)
- Professional mastery requires extended real-world practice

Part 2: Detailed Training Plan (Step-by-Step, Time-Based)

Assumptions

- Each course takes 2–4 days
- Average effort: 4–6 hours/day
- Training pace: 1 course per week
- Total timeline below focuses on initial readiness (approx. 13 weeks)
- Full mastery continues beyond this with practice

Phase 1: Foundation, Shadow Analyst & Bridge Setup (Weeks 1–3)

Week	Course	Time	Goal
1	Course 1: Note taking & Shadow Analyst for C Suite	2–3 days	Learn executive-level note taking, shadowing techniques, and how to support C-suite decision-making with structured observations
2	Course 2: Remote & Independent Working	2–3 days	Learn how to work independently; understand ticket-based and manager-less environments; set expectations for self-driven technical work
3	Course 3: Ubuntu Bridge Computers for Remote Work	3–4 days	Set up and manage Ubuntu-based bridge machines; understand networking, SSH, and remote connectivity essential for distributed infrastructure

Phase 2: Infrastructure & Core Python (Weeks 4–5)

Week	Course	Time	Goal
4	Course 4: Big Data Infrastructure & Ticket-Based Learning	3–4 days	Understand big-data systems; learn production-style workflows; exposure to infrastructure issues and troubleshooting
5	Course 5: Python Full-Stack Computational Framework (Beginner)	3–4 days	Python fundamentals; data structures and computational thinking; writing clean, reusable Python code

Phase 3: Testing, Research Writing & Intermediate Python (Weeks 6–9)

Week	Course	Time	Goal
6	Course 6: Comprehensive Testing & Error Reading	2–3 days	Debugging skills; understanding stack traces; reading and fixing errors independently
7	Course 7: Research & Writing in Controlled Environment	2–3 days	Structured documentation, clear technical writing, and research skills for controlled big-data environments
8	Course 8: Python Big Data Refresher (Intermediate)	3–4 days	Reinforce Python concepts; apply Python to data-heavy problems; build confidence in writing intermediate-level code
9	Course 9: Python Backend & ML Engines	3–4 days	Backend computation; understanding Python in ML / analytics pipelines; performance-oriented programming

Phase 4: Testing at Scale, Simulation Code & Distributed Systems (Weeks 10–13)

Week	Course	Time	Goal
10	Course 10: Writing Tests for Simulation & Code Conversion	2–3 days	Writing tests for complex logic; ensuring correctness during refactoring; quality control mindset
11	Course 11: Full-Stack Computational Framework for Big Data	3–4 days	End-to-end computation pipelines; simulation-based design; scaling logic from small to large systems
12	Course 12: Big Data Code Changes for Full-Stack Simulation Engine 2026	3–4 days	Hands-on modifying simulation engine code for big-data scale; real-world code change management
13	Course 13: Converting Standalone Code to Distributed Code	3–4 days	Distributed computing concepts; parallel execution; production-grade analytics workflows

Summary Timeline (Updated)

Phase	Duration
Foundation, Shadow Analyst, Remote Work & Bridge Setup	3 weeks
Core Python & Infrastructure	2 weeks
Testing, Research Writing & Intermediate Python	4 weeks
Distributed Systems, Simulation Code & Scale	4 weeks
Initial Readiness	13 weeks
Professional Mastery	6–24 months with practice

How to Use This Plan

- Follow strict order
- Do hands-on coding alongside videos
- Re-do exercises if unclear
- Expect difficulty — that's normal
- This is career-level training, not casual learning

Appendix: Expanded Acronyms & Pedagogical Basis

Expanded Acronyms (Full Forms)

- **ML** — Machine Learning
- **MC** — Monte Carlo (simulation methods)
- **SimEng** — Simulation Engineering
- **Infra** — Infrastructure
- **Admin** — Administration
- **Python** — Not an acronym; a programming language named after Monty Python
- **API** — Application Programming Interface
- **CI/CD** — Continuous Integration / Continuous Deployment
- **SSH** — Secure Shell
- **Ubuntu Bridge** — A computer acting as a network bridge/gateway for remote distributed systems

Why This Strict Order? (Pedagogical Explanation — Updated for 13 Courses)

The training plan follows a **cumulative skill ladder**.

1. **Week 1 (Shadow Analyst)** — Executive communication and observation skills; foundation for understanding business context.
2. **Week 2 (Remote Work)** — Metacognitive foundation: how to learn independently.
3. **Week 3 (Ubuntu Bridge Computers)** — Before any cloud or big-data infra, you must know how to set up and manage the actual bridge machine (Ubuntu) that connects remote work environments. Essential for the "bridge role."
4. **Week 4 (Big Data Infrastructure)** — Understand ecosystems (clusters, pipelines) before writing code.
5. **Week 5 (Python Beginner)** — Core programming fundamentals.
6. **Week 6 (Testing & Error Reading)** — Debugging taught early for self-correction.
7. **Week 7 (Research & Writing)** — Structured documentation and controlled-environment research skills. Critical for tickets and simulation reports.
8. **Week 8 (Python Intermediate Refresher)** — Reinforcement through quiz-based, data-heavy problems.
9. **Week 9 (Backend & ML Engines)** — Performance-oriented Python for real pipelines.

10. **Week 10 (Writing Tests)** — Non-negotiable for distributed systems.
11. **Week 11 (Full-Stack Big Data Framework)** — Integrates all previous skills.
12. **Week 12 (Big Data Code Changes for Simulation Engine)** — Real-world modification of existing simulation engine code. Teaches how to safely introduce big-data changes without breaking simulations.
13. **Week 13 (Standalone → Distributed)** — Capstone: convert single-machine code to parallel, distributed execution.

Cognitive Load Management

- Early weeks focus on *one new major concept* per week.
- Shadow analyst skills establish business communication before technical depth.
- Ubuntu bridge skills are introduced immediately after remote work — practical before theoretical infrastructure.
- Research/writing is placed after testing to reinforce documentation of errors and test results.
- Big-data code changes come before final conversion to distributed code — you first learn to modify existing engines, then redesign from standalone to distributed.

Time-to-Competence Realism (Updated for 13 Courses)

- **13 weeks** → Initial readiness (can complete supervised tickets, understands bridge role basics)
- **6 months** → Consistent independent work on medium-complexity tasks
- **1–2 years** → Full mastery (architecting distributed solutions, mentoring others)

Document generated from the original training plan by Shivgan Joshi. All Udemy links are preserved as provided. Updated with all 13 courses in strict sequential order.